INCH-POUND

MIL-C-83513/11A
28 June 1993
SUPERSEDING
MIL-C-83513/11
11 February 1992

MILITARY SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, RECTANGULAR, PLUG, MICROMINIATURE, POLARIZED SHELL, RIGHT ANGLE, PIN CONTACTS, 3 ROW ,SOLDER TYPE, NARROW PROFILE, 51 CONTACTS, PRINTED CIRCUIT BOARD

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification Listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation MIL-C-83513

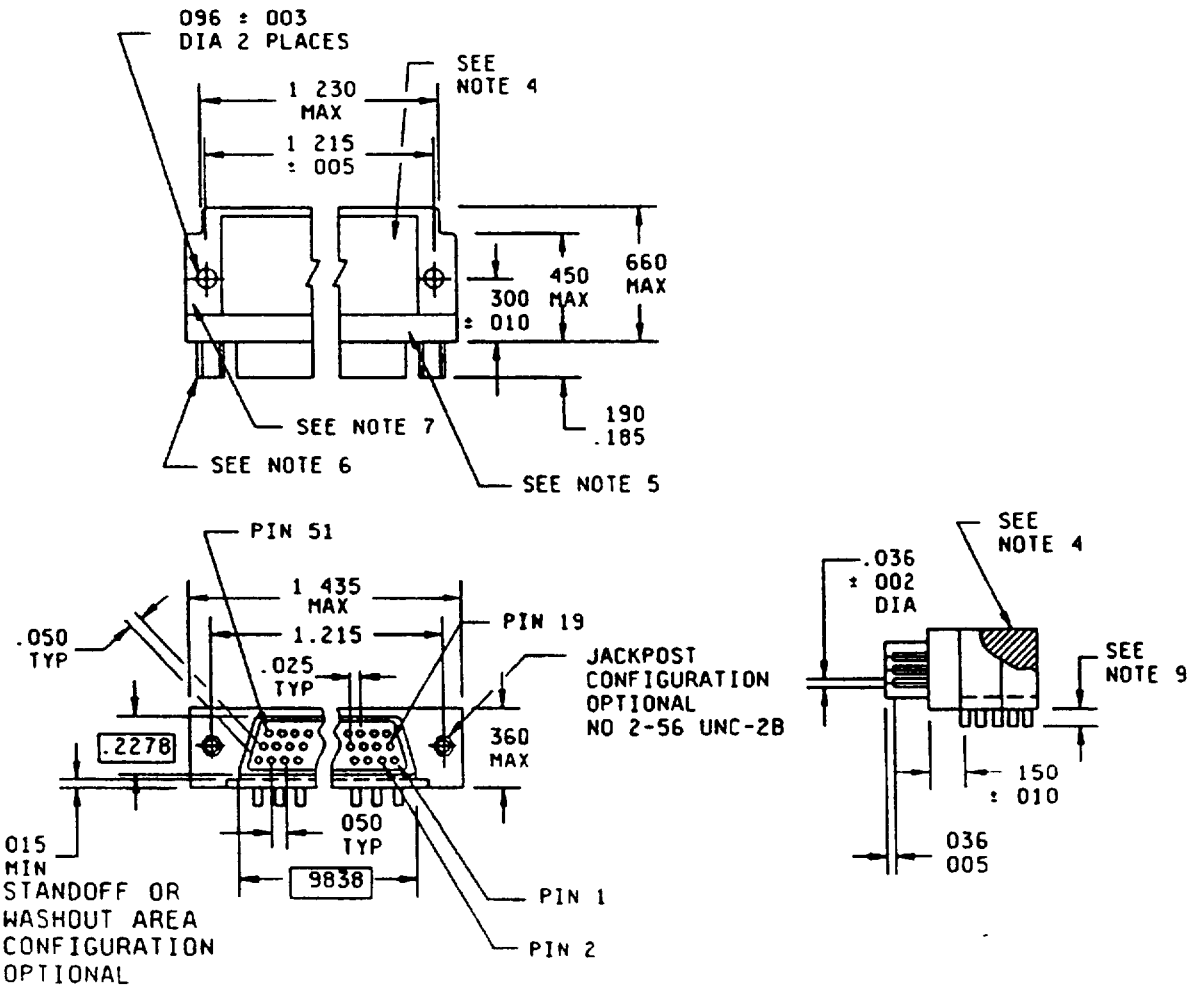
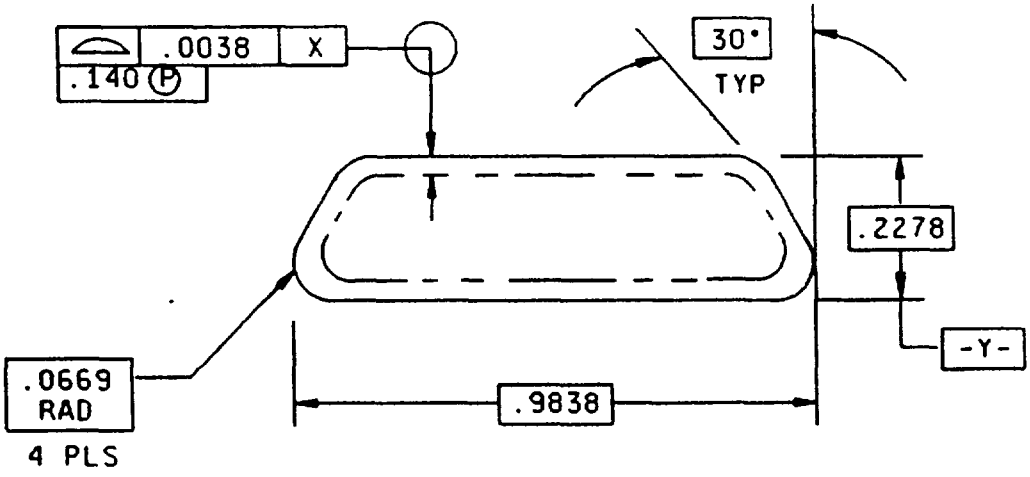


FIGURE 1 Connector, plug, 050 spacing

DISTRIBUTION STATEMENT A Approved for public release, distribution is unlimited

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:28Z
Check the source to verify that this is the current version before use.



DETAIL OF INTERFACE

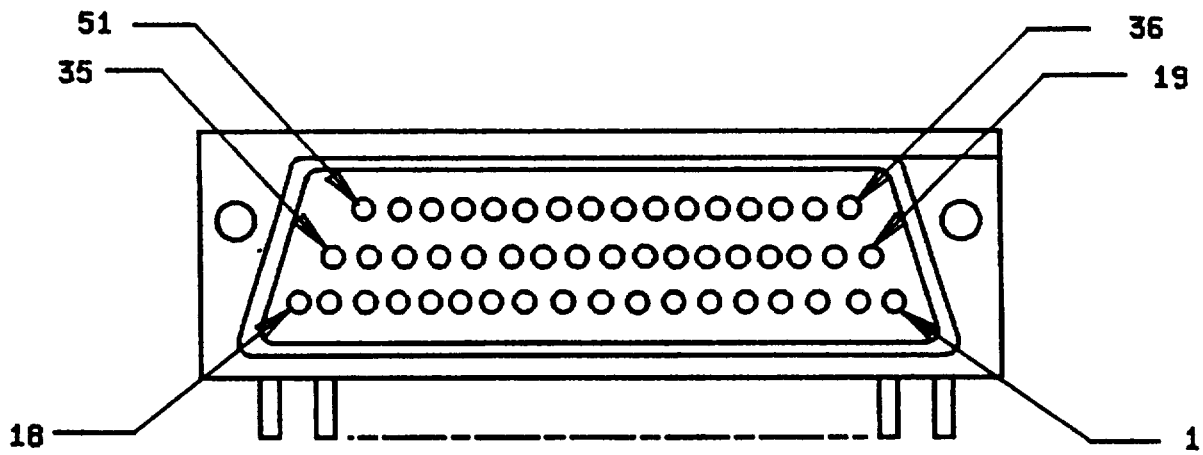
Inches	mm	Inches	mm
002	0 05	185	4 70
.003	0.08	190	4 80
010	0.25	.2278	5.786
.015	0 38	.300	7.62
025	0 64	360	9.14
.036	0 91	.450	11 44
.050	1.27	660	16.76
.056	1 42	.9838	24 989
.0669	1 699	1.215	30.86
.096	2.44	1.230	31 24
109	2 77	1 435	36 45
150	3.81		

NOTES

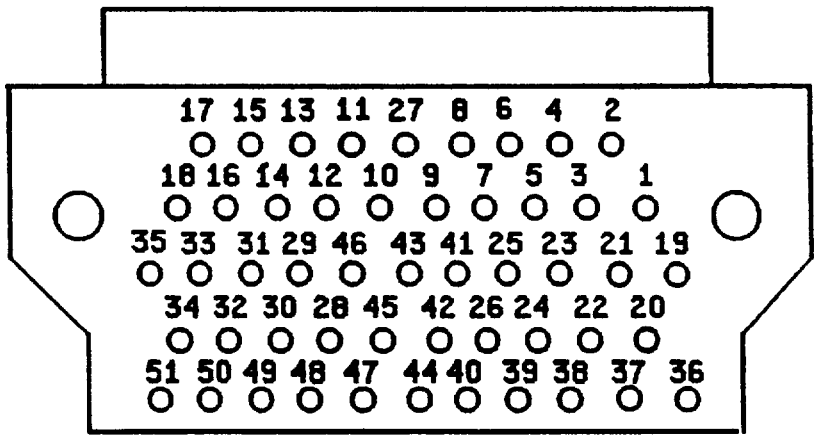
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances are ±.005 (0.13 mm).
4. Termination organization area to be optionally molded or filled with a potting fill material capable of passing the electrical and environmental requirements of MIL-C-83513 Plastic molding shall conform to the requirements of GDI-30F or type SDG-F of MIL-M-14, or GST-40F or GLCP-30F or GLCP-50 of MIL-M-24519
5. Metal shell shall be of material in accordance with MIL-C-83513 for class M parts.
6. Jackpost (permanently attached), when specified Corrosion resistant steel in accordance with ASTM A484 and ASTM A582, 300 series stainless steel, passivated in accordance with QQ-P-35
7. Separately molded plastic body (if used) shall conform to the requirements of type GDI-30F or type SDG-F of MIL-M-14, or GST-40F or GLCP-30F or GLCP-50 of MIL-M-24519
8. Wire termination pins shall conform to QQ-W-343, no 24 AWG copper
9. Termination lengths available 109 (2 77 mm), 140 (3 56 mm) or 172 (4 37 mm) The tolerance shall be ± 015 (0 381 mm) for all termination lengths

FIGURE 1 Connector, plug, 050 spacing - Continued

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:28Z
Check the source to verify that this is the current version before use.



51 CONTACT PLUG PIN CONNECTOR
- RIGHT ANGLE -

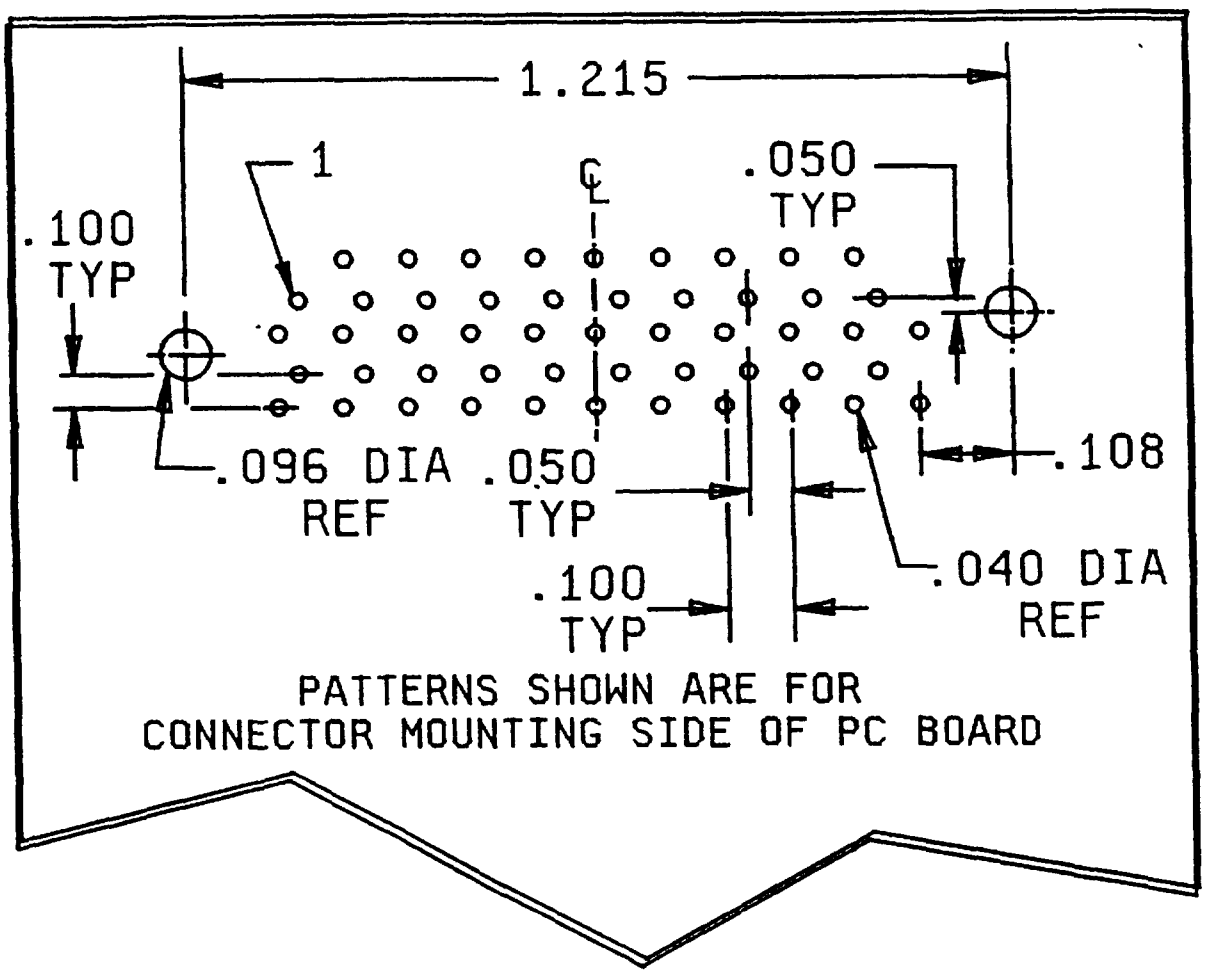


TERMINATION VIEW - RIGHT ANGLE
NARROW PROFILE

NOTE Engaging face of pin insert shown, cavity identification numbers are for reference only and do not appear on the part

FIGURE 2 Insert arrangement

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:28Z
Check the source to verify that this is the current version before use.



Inches	mm
.040	1.02
.050	1.27
.100	2.54
.108	2 74
125	3 18
1 215	30 86

NOTES:

1. Dimensions are in inches
2. Metric equivalents are given for information only
3. Unless otherwise specified, tolerances are ± 005 (0 13 mm)

FIGURE 3 Layout arrangement

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:28Z
Check the source to verify that this is the current version before use.

REQUIREMENTS

Dimensions and configurations. See figures 1 and 2

Current rating, maximum 3 amperes per contact

Materials

Termination organization area Potting fill material capable of passing the electrical and environmental requirements of MIL-C-83513

Shell. The requirements for shell materials shall be in accordance with MIL-C-83513

Plastic body or plastic molding Shall conform to the requirements of GDI-30F or type SDG-F of MIL-M-14, or GST-40F or GLCP-30F or GLCP-50 of MIL-M-24519

Jackpost. Corrosion resistant steel in accordance with ASTM A484 and ASTM A582, 300 series stainless steel, passivated in accordance with QQ-P-35

Wire termination pins Wire termination pins shall conform to QQ-W-343, no 24 AWG copper.

Mating connector: Shall conform to MIL-C-83513/2 and MIL-C-83513/4

Plating of termination leads Solder dipping of termination leads will be accomplished in SN60 PB40 or SN63 PB37 in accordance with QQ-S-571

Part or Identifying Number (PIN). The PIN shall consists of the letter M, the basic number of the specification sheet, a letter for the insert arrangement, a numerical code for the termination length, and a letter code from the shell finish and hardware columns

M83513/11-	G	01	C	P
Specification sheet number	Insert arrangements (see figure 2) G = 51	Termination length 01 = 109 02 = 140 03 = 172	Shell finish C = Cadmium N = Electroless nickel (space application)	Hardware N = No jackpost P = Jackpost attached

Revision letters are not used to denote changes due to the extensiveness of the changes

CONCLUDING MATERIAL

Custodians
Army - CR
Navy - EC
Air Force - 85
NASA - NA
Review activities
Army - MI
Navy - AS, SH
Air Force - 11, 17, 99
DLA - ES

User activities
Army - AT, ME
Navy - CG, MC

Preparing activity
Air Force - 85

Agent
DLA - ES

(Project 5935-3921-02)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T08:28Z
Check the source to verify that this is the current version before use.